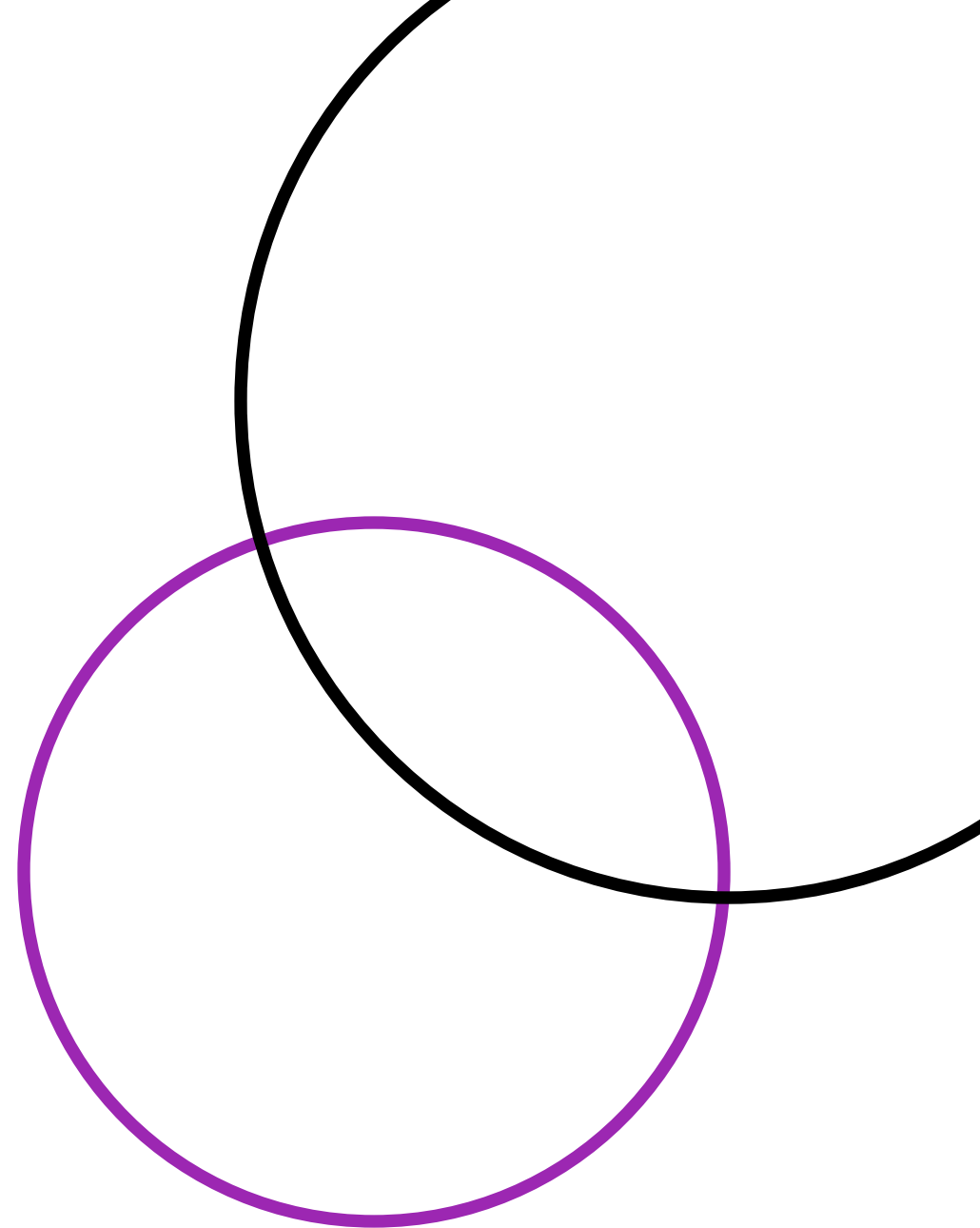




wellcome
connecting
science

MPRAs: Exploring and QC-ing non-coding variation with MAVEs

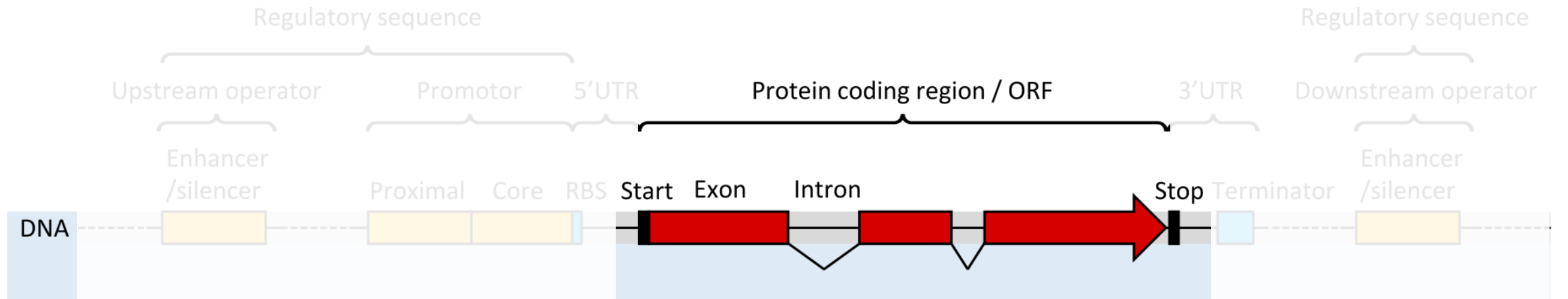
A/Prof Irene Gallego Romero
St Vincent's Institute of Medical Research



Intended learning outcomes for the module

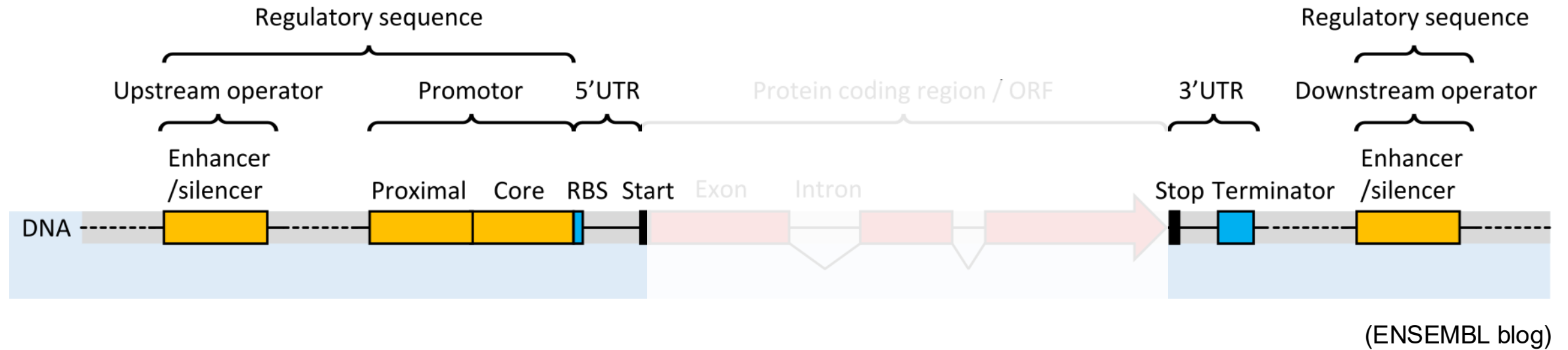
1. Discuss the importance of robust study design and calibration in MPRAs and ways in which they differ from other MAVEs.
2. Perform QC on datasets from different assays.
3. Determine which QC steps need to be completed for different datasets and interpret the results of each step.

The course so far:



(ENSEMBL blog)

MPRAs:

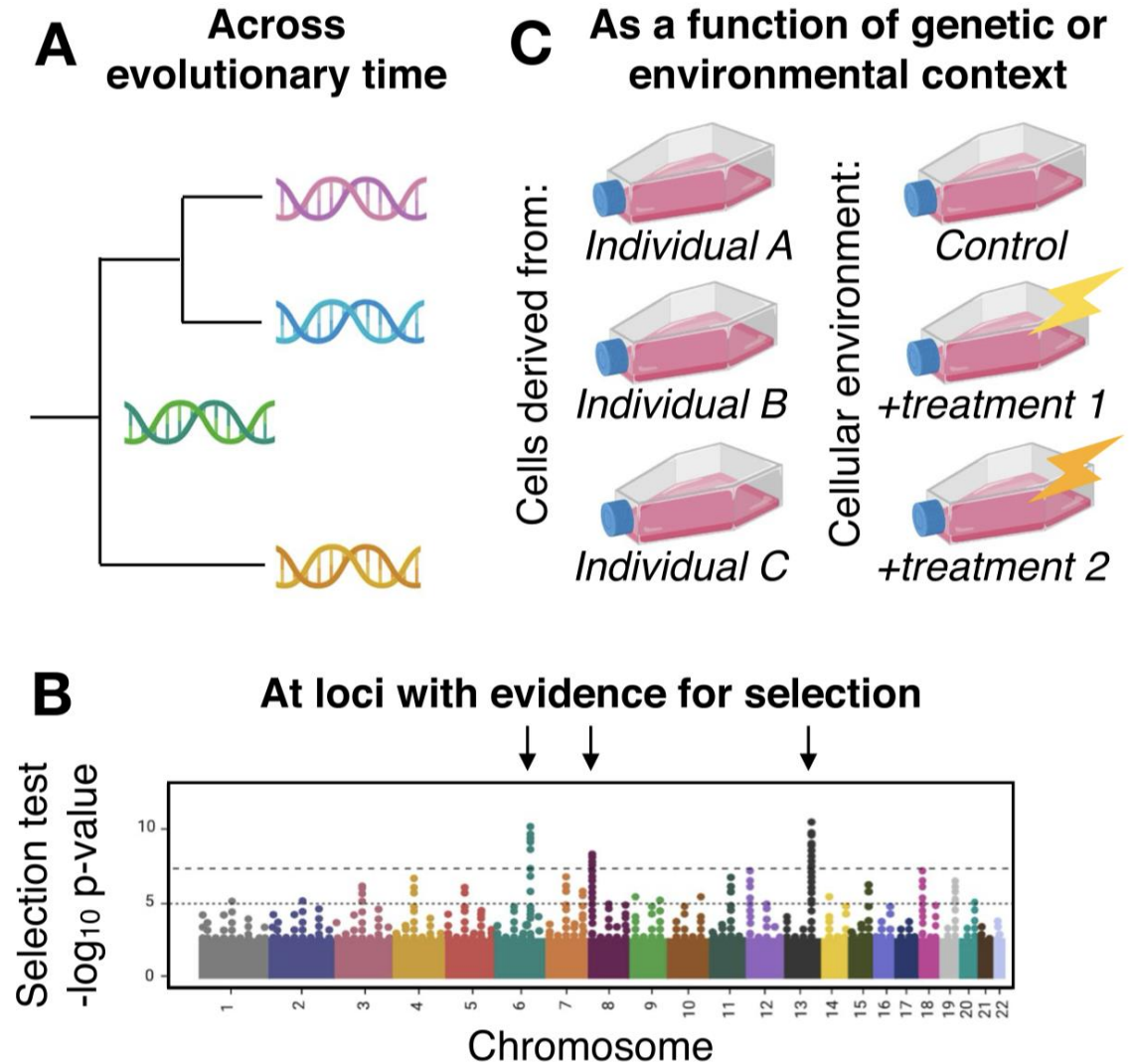


But why?

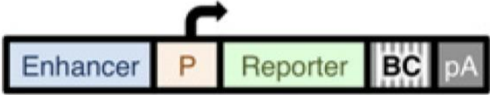
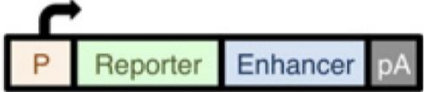
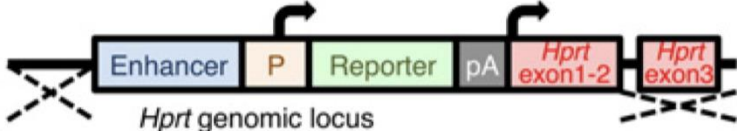
- Non-coding regions contribute to traits, although individual effect sizes are generally small compared to coding regions
 - Effect sizes means they tend to contribute more to complex, rather than monogenic/Mendelian, traits
 - This requires special consideration for study design
- Most MPRA test many genomic elements, rather than a single element or region in high depth
 - Move away from DMS/SGE paradigm
 - Also requires consideration in study design
- (They're more fun than coding regions)

And what?

- Dissect single enhancer or promoter
 - Same DMS approach, different target, different read out
- Evolutionary questions broadly
 - Traversing a fitness landscape
 - Examine ancestral states
- Fine mapping of GWAS or eQTL peaks
 - Identify causal variants, dissect biological mechanism
- Test a selection of regions genome-wide for function
 - Test the same element in many genes?
 - Test specific variants to identify allelic effects?
- Often, combinations of all of these



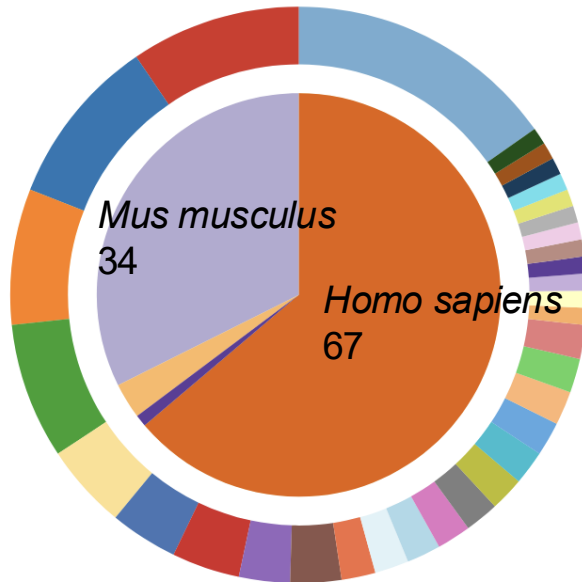
MPRA flavours

	Construct	Tested in	Detection	Advantages	Disadvantages	References
MPRA/ MPFD/ CRE-seq		Cell lines, Mouse liver, Mouse retina	Barcode RNA-seq	High BC multiplicity Quantitative	Episomal	2, 24-29, 36, 38
STARR-seq		Cell lines	Enhancer RNA-seq	Quantitative	Low multiplicity Episomal	30-32
TRIP		Mouse ESCs	Barcode RNA-seq	Quantitative Genomic context	Low resolution	33
FACS-based		Yeast	FACS	Semi-quantitative	Low multiplicity Episomal	39
FACS-based (FIREWACH)		Mouse ESCs	FACS	Chromosomal	Not quantitative	34
FACS-based (SIF-seq)		Mouse ESCs	FACS	Chromosomal	Not quantitative	35

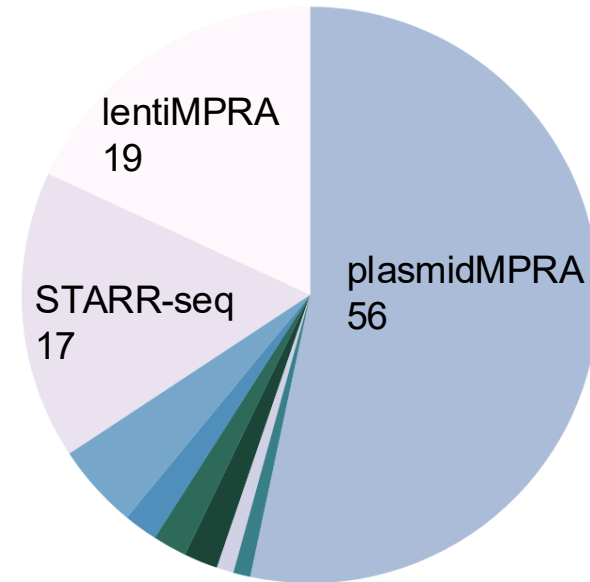
MPRAbase

- <https://mprabase.ucsf.edu/>
- Not quite as functional as MaveDB... but the data is thornier!
- Summary of studies, but not interactive. Nonetheless gives a good overview of the field

Species / Cell Lines



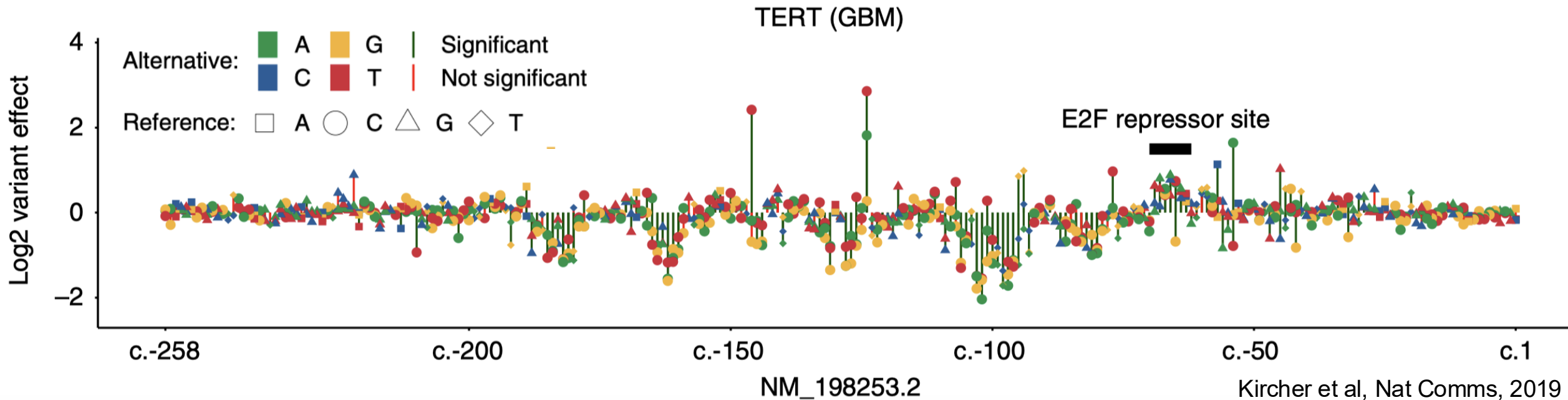
Library Strategy



MPRA flavours (I)

Dissecting a single enhancer through DSM:

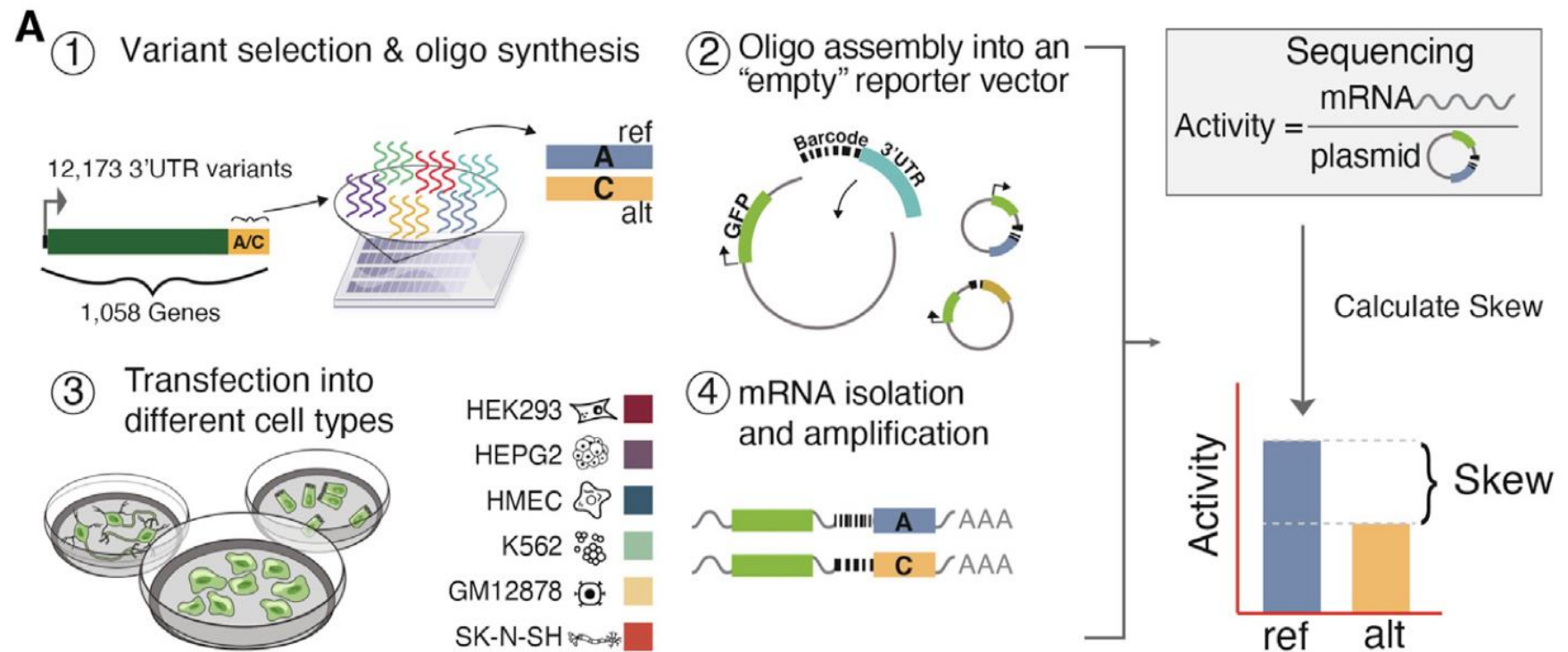
- "Original" MPRA (Patwardhan et al, at Biotech, 2009) but not a common approach (only 1 reported in MPRAbase)
- Build saturation maps of specific target region
 - Easy in model organisms with small genomes, harder in humans once you go far beyond proximal enhancers



MPRA flavours (II)

Test a selection of regions genome-wide for function

- Assess context specificity or dissect functional mechanism of a particular genetic feature
 - Works better when you can identify those regions easily...



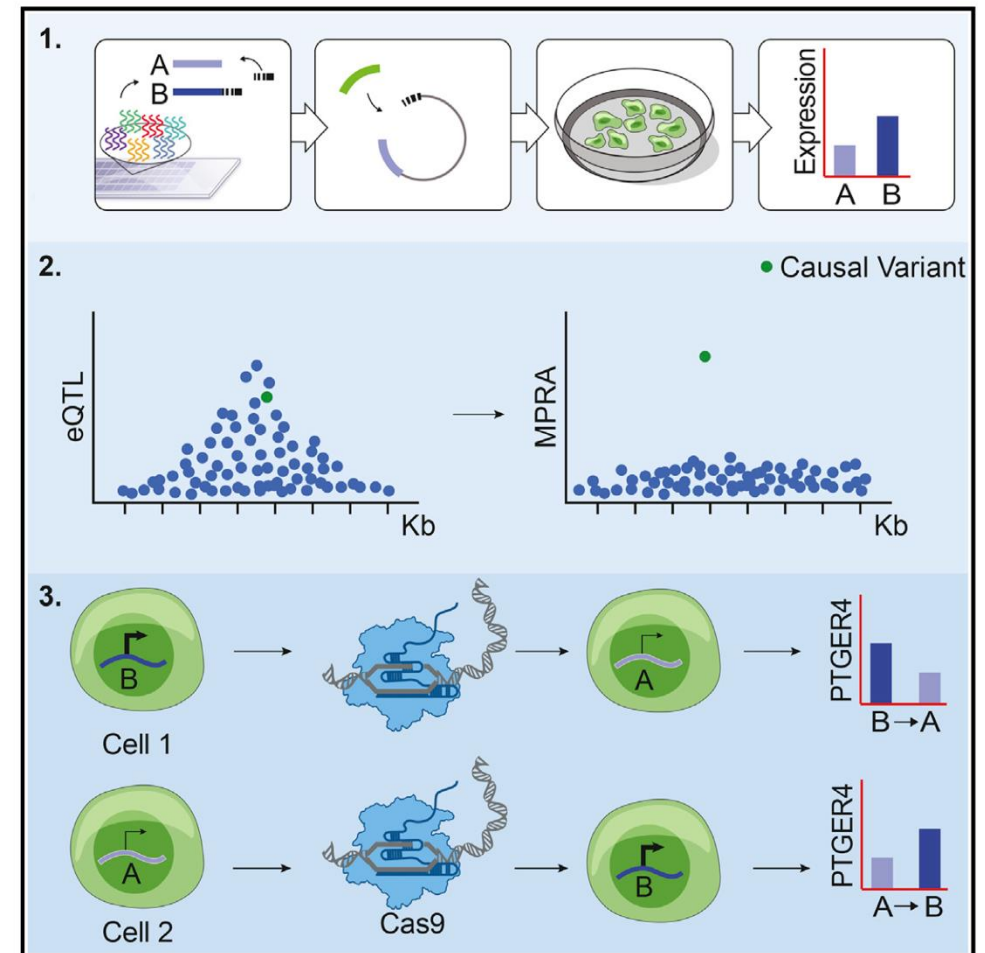
MPRA flavours (III)

Fine-mapping a GWAS/eQTL peak in high LD

- Identify causal variants from a set of high-LD variants flagged by association mapping
 - Complement to existing computational approaches for fine-mapping and colocalisation
 - Can combine with cell perturbations to identify dynamic drivers of gene regulation
 - Test many sites to identify **allele-specific** effects
 - Assume limited number of causal variants? Expect a small number of significant hits per tested peak?

Tewhey et al, Cell, 2016

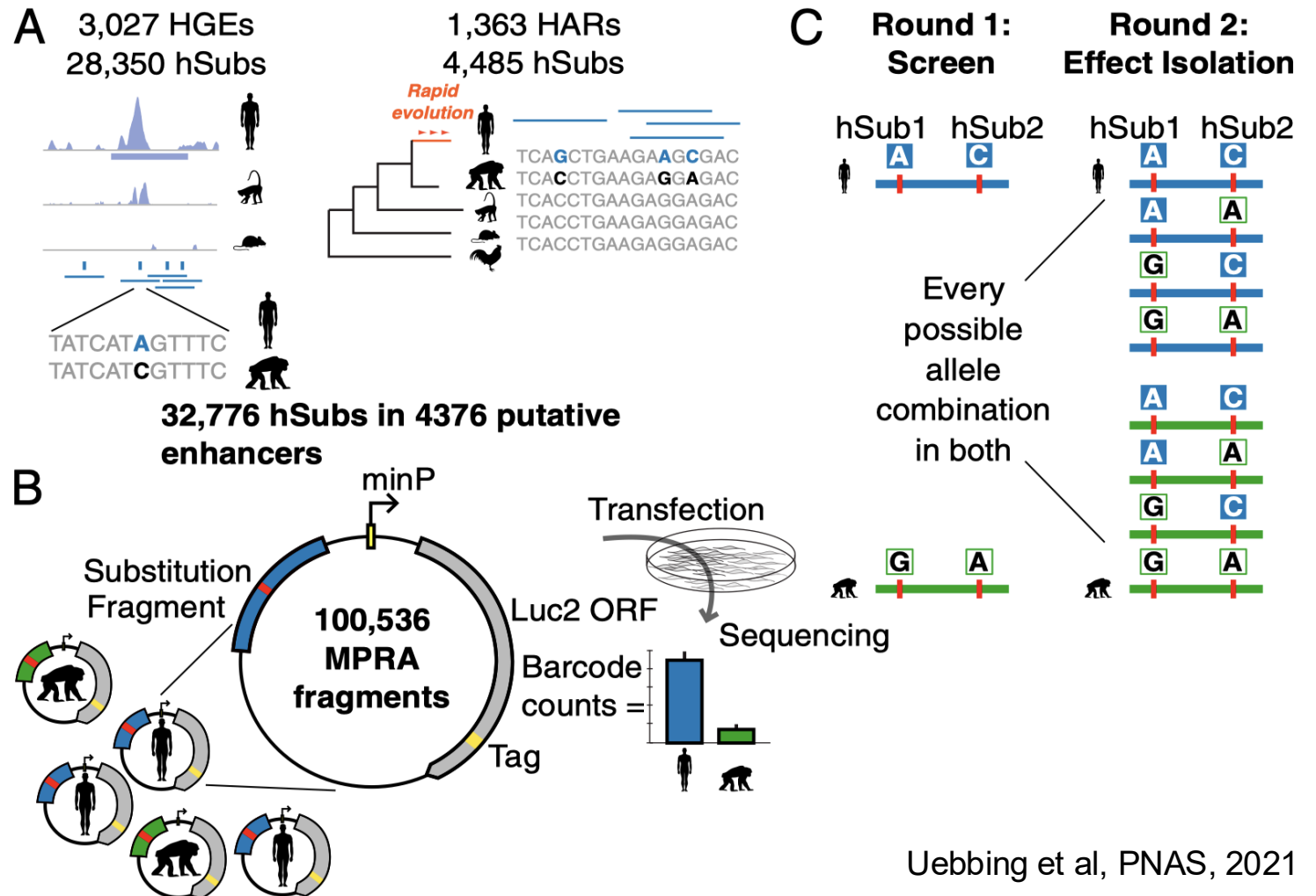
Graphical Abstract



MPRA flavours (IV)

Comparing a set of SNPs genome-wide

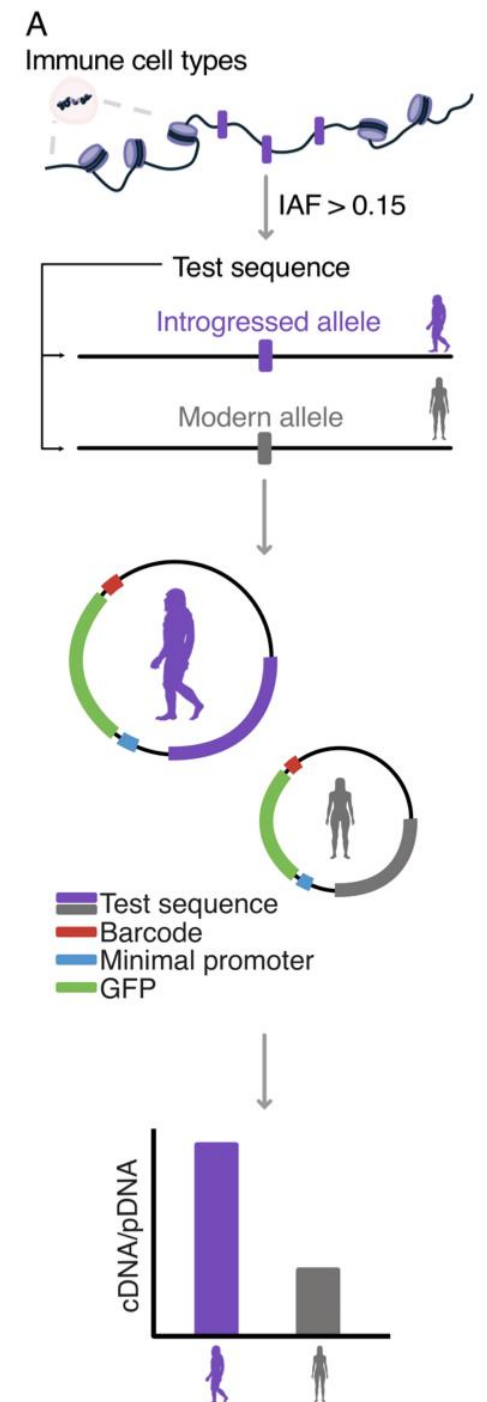
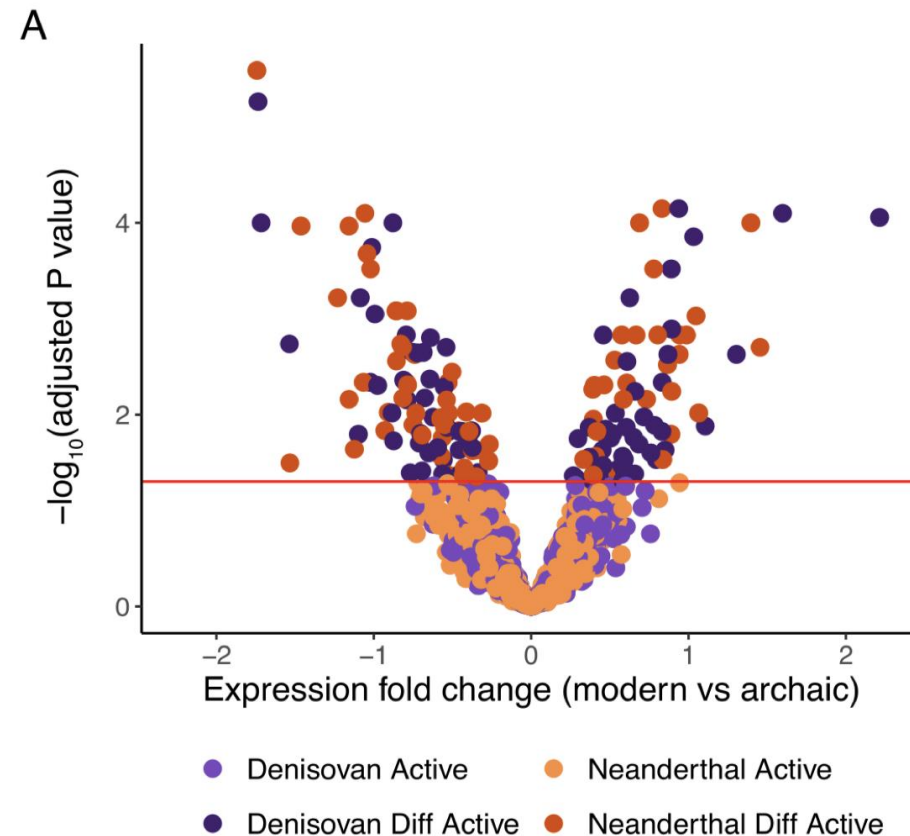
- Like III, but without the assumptions of LD – variants are drawn from the whole genome on the basis of some interesting feature
 - Evolutionary state?
 - Genomic location?
- Generally interested in identifying *allelic-specific* effects.
 - Analysis involves many pairwise comparisons of independent loci



MPRA flavours (IV)

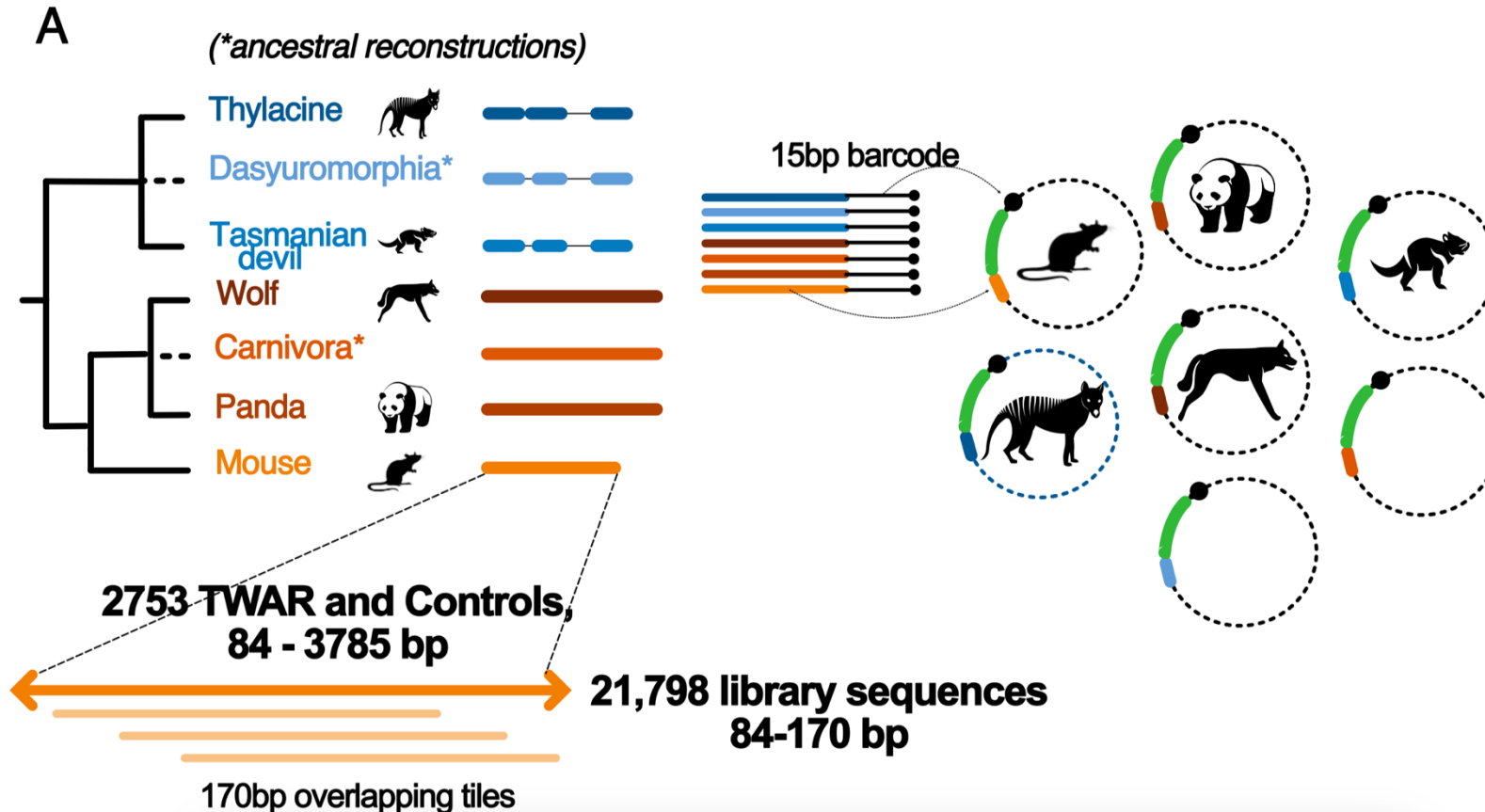
Comparing a set of SNPs genome-wide

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MPRA flavours (V)

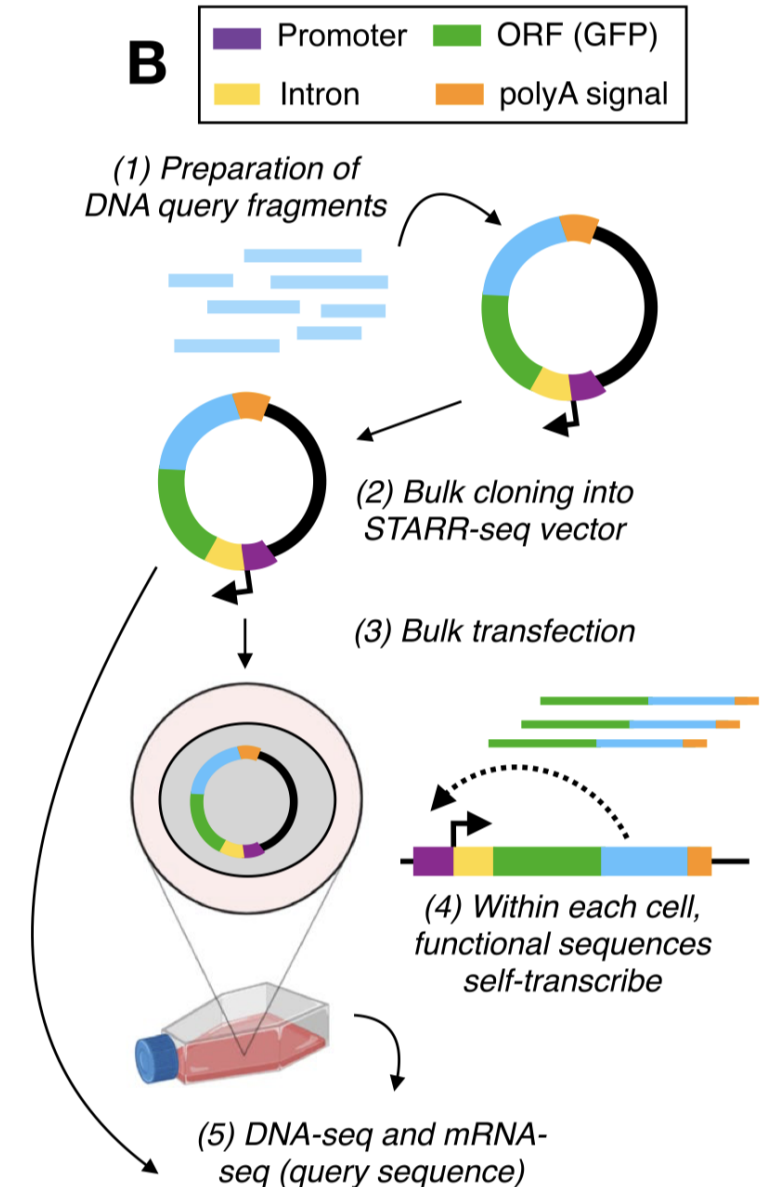
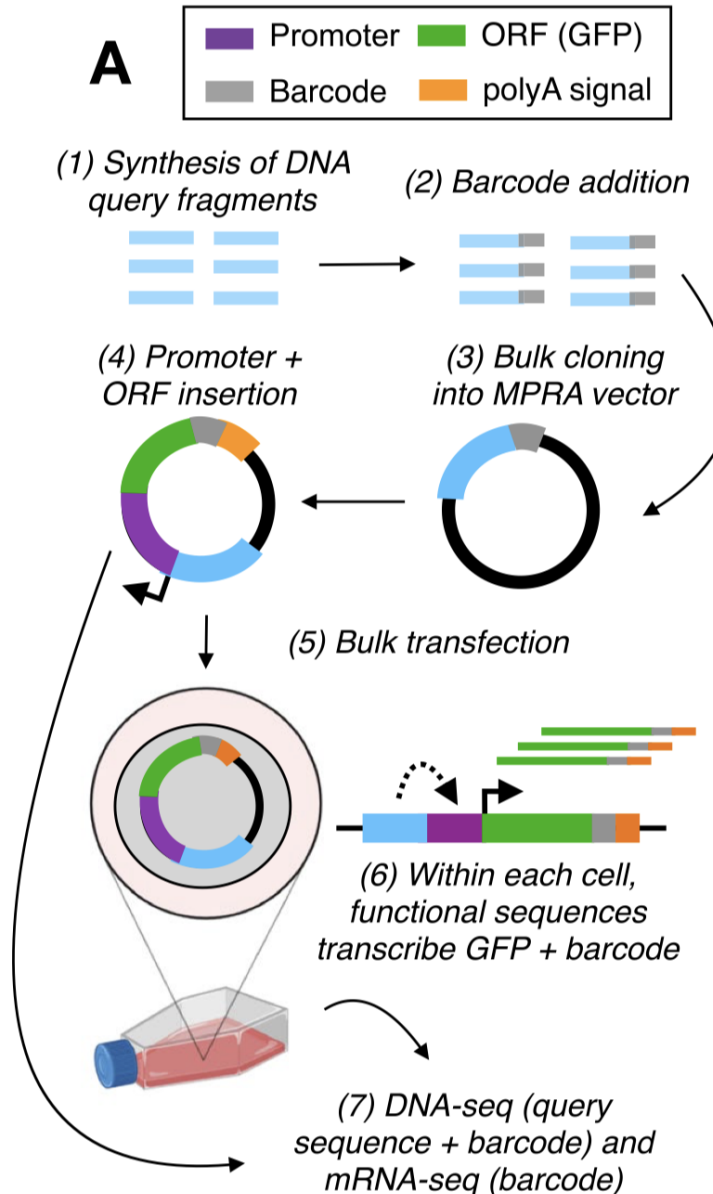
Testing evolution of elements across deep time



- More complex designs also possible:
 - Comparisons across phylogenetic clades, including reconstructed ancestral states
- Sequence turnover and element size present challenges
 - Single regulatory elements are generally larger than what fits into a single plasmid

Generic MPRA workflows:

- MPRAs plasmids are generally either lentiviral (integrating) or episomal (non-integrating).
 - Neither of these mimics the true chromatin state of the element being tested
- "Classic" MPRA design:
 - Target element is upstream of minimal promoter + reporter + barcode
 - Test regions generated through oligosynthesis
 - Barcodes added through degenerate PCR
- STARR-seq design:
 - Target element is downstream of minimal promoter
 - Element acts as its own barcode + reporter
 - Can use output of ChIP-seq, ATAC-seq, etc...



Hasty notes on experimental design:

- Number of elements:
 - Need to maintain library complexity throughout, so the higher the number the bigger the experiment – 50-60,000 unique oligos in a library is near the top of the advised range. If $>$, split into smaller libraries.
- Number of barcodes:
 - Aim for 80-200 at the library association step. If you go much higher, you might want to consider subculturing the library, as otherwise required sequencing depth to link barcodes and oligos will be needlessly high.
- Number of replicates:
 - 2-3 for identifying activity, 4-6 (Myint et al, 2019) or more for detecting allelic effects.
- Number of cells:
 - Assume $MOI > 1$ but keep in mind barcode representation and transfection efficiency. Some people are starting to incorporate UMIs.
 - For ~50k library, ≥ 50 million cells is good, but extremely contingent on cell line.

Lentiviral or episomal?

Lentiviral	Episomal
• Integrates into the genome (but no control over where)	• Non-integrating
• Lower throughput	• Higher throughput
• Correlates better with "true" biological signal	• Less faithful to biological context, but more internally replicable due to lack of integration effects
• Stable expression (but can be silenced)	• Transient expression (easily lost or diluted especially if cells keep proliferating)
• Can be used in more cell types	• Many cells hate it.

Generic MPRA output:

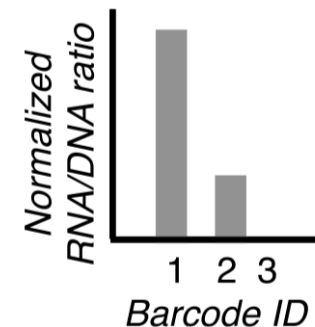
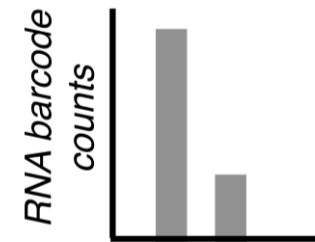
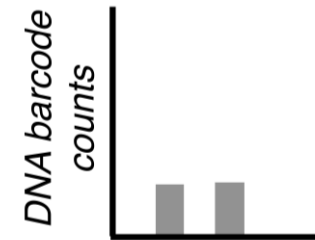
- Pre-transfection DNA-seq to quantify plasmid abundance can also be done post-transfection, to get a better idea of how the plasmid pool looked at the time of transfection
- Post-transfection, the reporter construct is used to do a pulldown and barcode RNA levels are quantified through RNA-seq
- RNA/DNA ratio is foundation of virtually all MPRA data analysis

Pre transfection
DNA-seq measures
experimental input

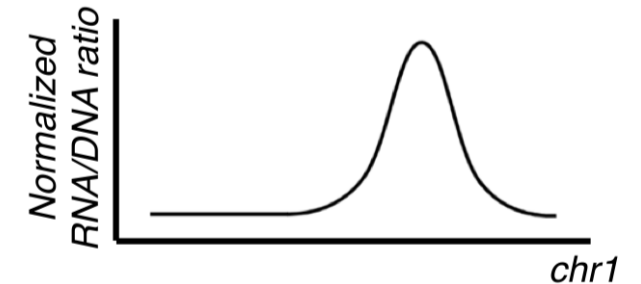
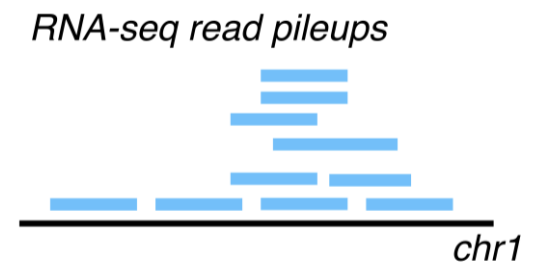
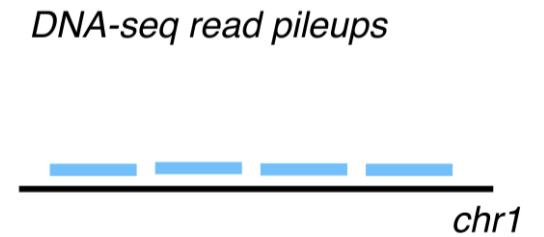
Post transfection
RNA-seq measures
regulatory output

RNA/DNA ratio
quantifies regulatory
activity

A Barcoded MPRA



B STARR-seq



Generic analysis steps:



- MPRAs are a **fussy and complex technology**, and most data analysis revolves around comprehensive data QC at different steps.
 - (We'll skip library design, but also challenging)
- Statistical modelling of MPRA data is a growing field and there's a lot of competing tools.
- Some of these steps can be combined/are handled simultaneously, depending on choice of method for testing.

Generic analysis steps:



- You can roll your own, but...
 - MPRAFlow (<https://mpraflo.readthedocs.io/en/latest/>)
 - MPRASnakeflow (<https://github.com/kircherlab/MPRASnakeflow>)
 - MPRAmatch (https://github.com/tewheylab/MPRA_oligo_barcode_pipeline)

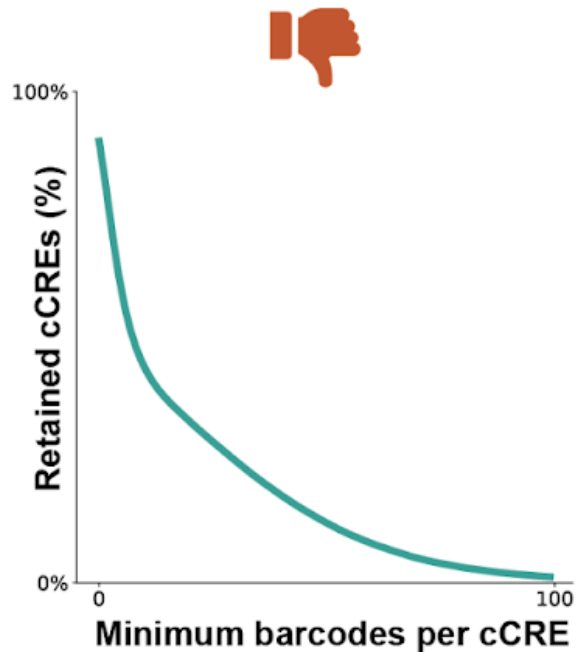
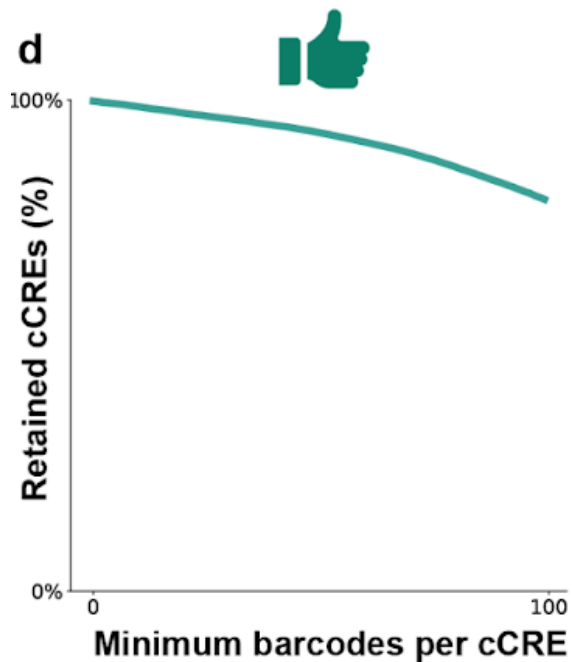
Generic analysis steps:



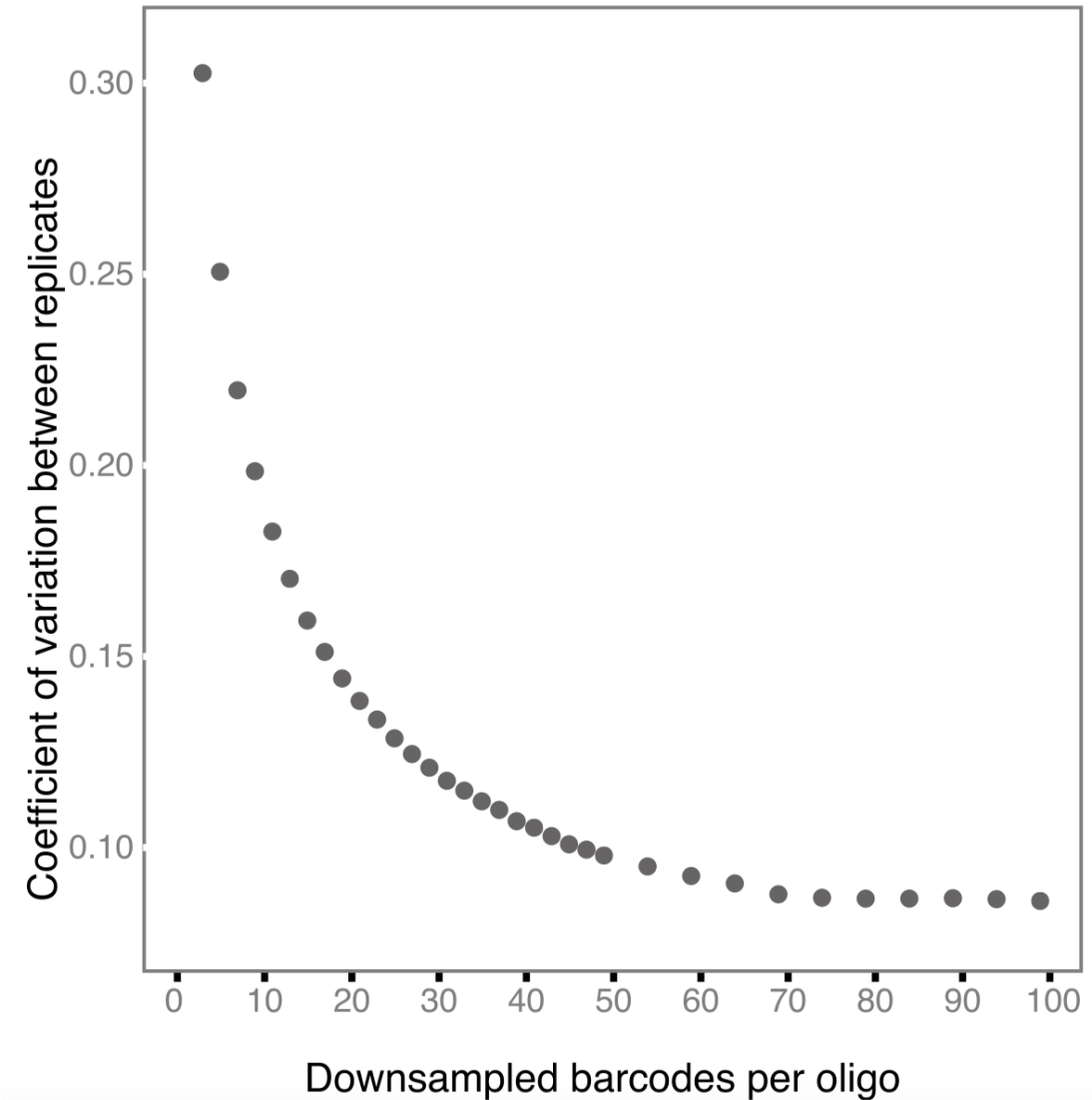
- Ensure ~even barcode representation
- Examine distribution of barcodes across and within oligos
 - (Identify (and remove???) outlier barcodes)
- Examine replicates at barcode level
- Check GC biases

How many barcodes?

- The more barcodes, the better protected you are against PCR jackpotting and outliers, but the deeper you have to sequence, and the more cells you need.
 - So where is the sweet spot?



Fishilevich et al, in prep.



Tewhey et al, Cell, 2016

Defining activity:

$$\log_2 \frac{RNA_{normalised\ counts}}{DNA_{normalised\ counts}}$$

- RNA (sometimes also referred to as cDNA) are the reads derived from transcription of the plasmid and capture transcriptional activity (and regulatory potential) of each tested element
- DNA (sometimes also referred to as pDNA) are reads directly derived from the plasmid and capture plasmid abundance in the transfected library

Generic analysis steps:

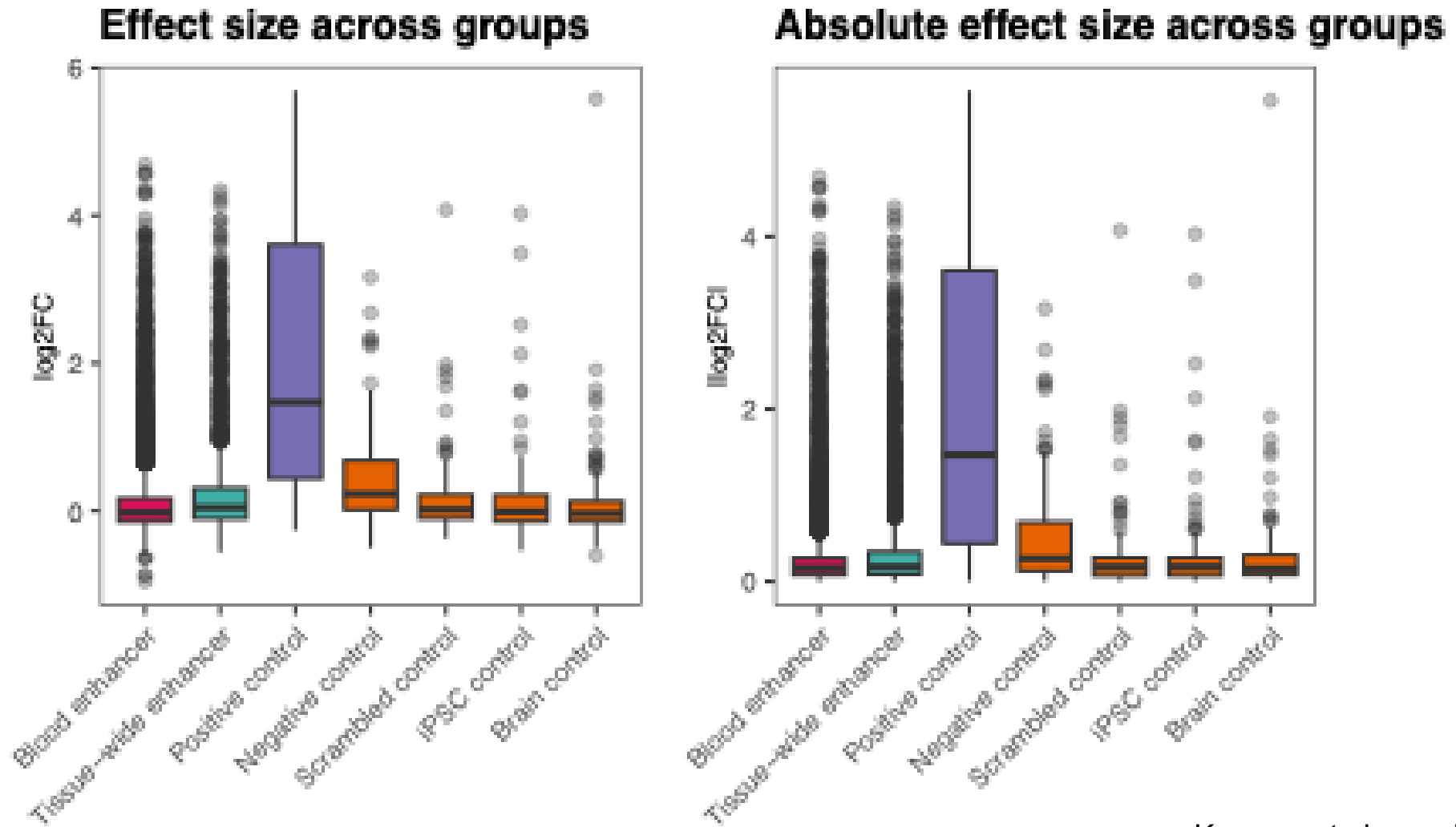


- Validate controls
- PCA
- Aggregate barcode information to oligo level
 - Why or why not?
- Assess reproducibility at the oligo level
- Check GC biases again

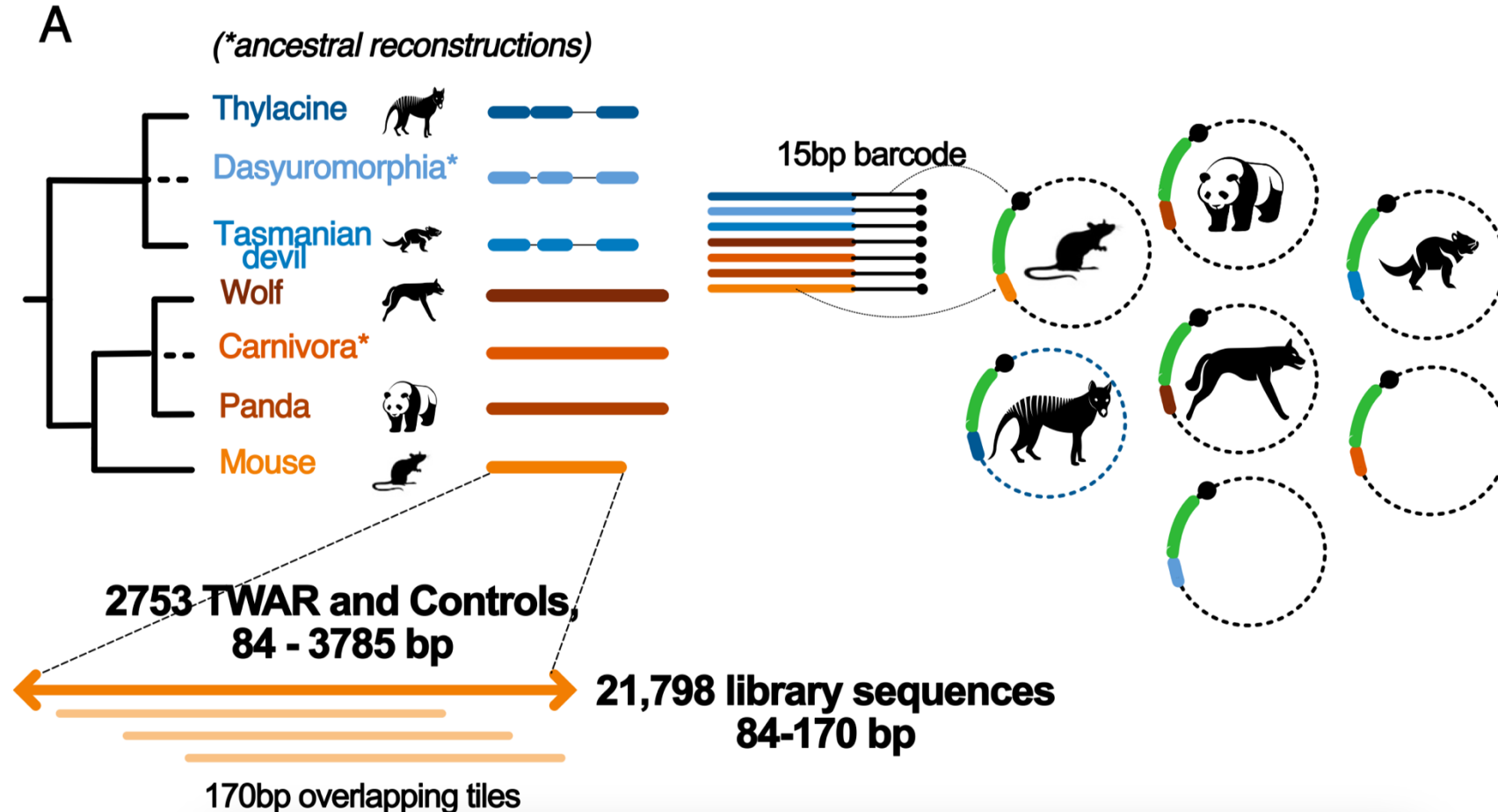
What makes a good control?

- $\neg_(\text{ツ})_/\neg$
- MPRAAs can be very hard to calibrate. Ideal positive control should be:
 - Previously validated ...
 - ... using the same technology...
 - ... in your cell type ...
 - ... but then if you have this you've already done this MPRA, so why are you repeating it?
- Negative controls
 - Elements that should not be active in your cell type (under the same conditions)
 - Scrambled positive controls (maintain GC content)
- The expected amount of separation between controls and test regions varies significantly depending on the type of region being tested. Designing robust controls is hard!

A successful experiment (?)



What are the controls in this experiment?



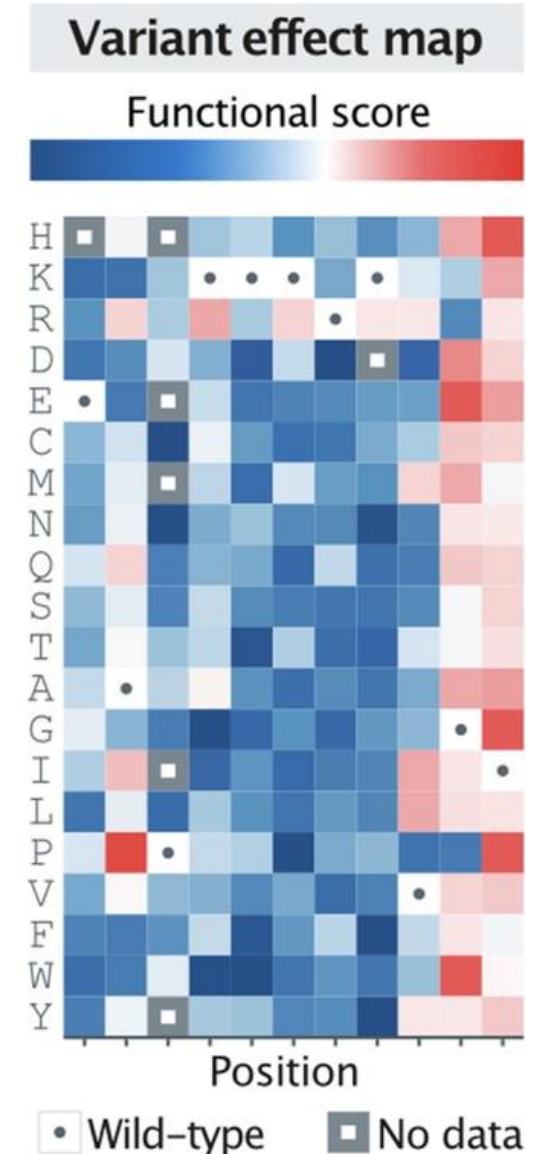
Generic analysis steps:



- Depending on study design, this might be last step (e.g. identify active enhancers in a large pool)
- But first you must define "active"!
- Define linear model and decide on statistical approach
 - Barcode level vs oligo level?
 - Use $\log_2$ RNA/DNA, or keep materials separate?

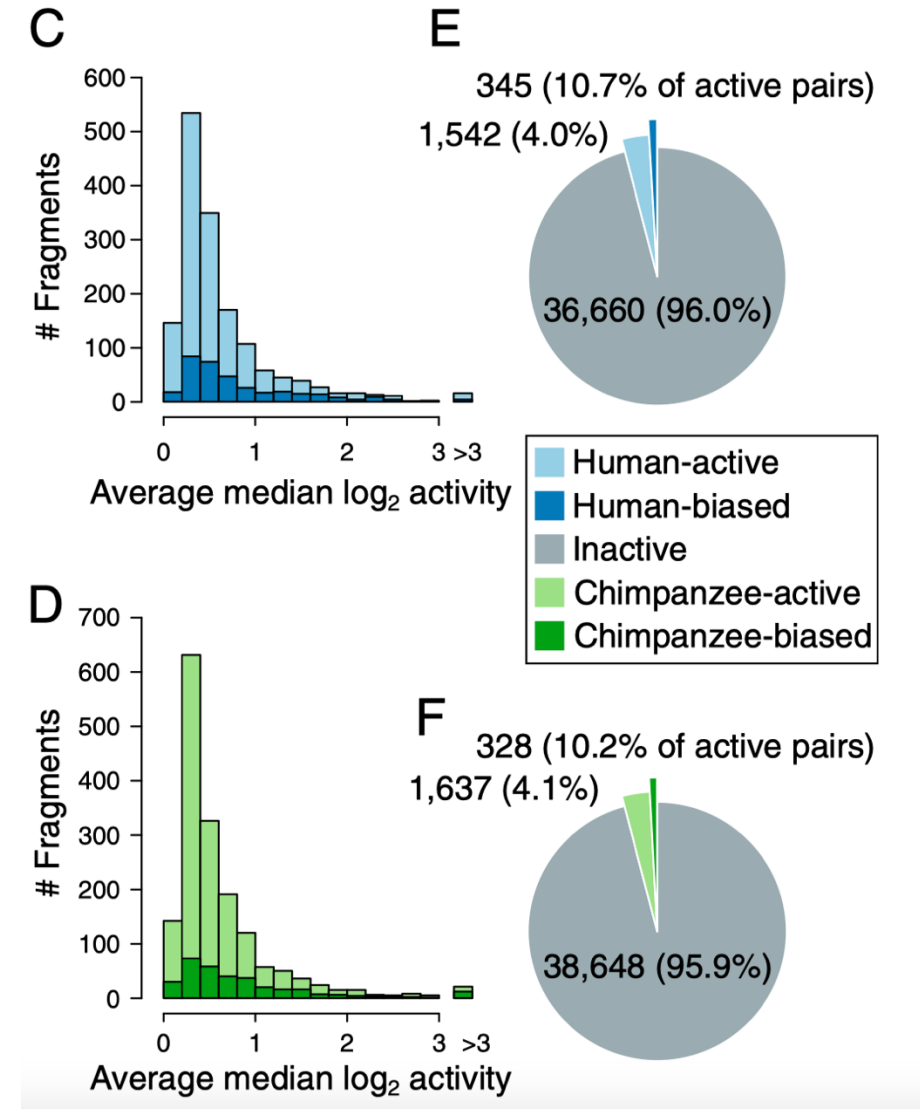
The challenge of defining activity

- Protein-coding DMS and SGE generally have inbuilt controls – wildtype sequence is expected to be 'active' and can be treated as baseline



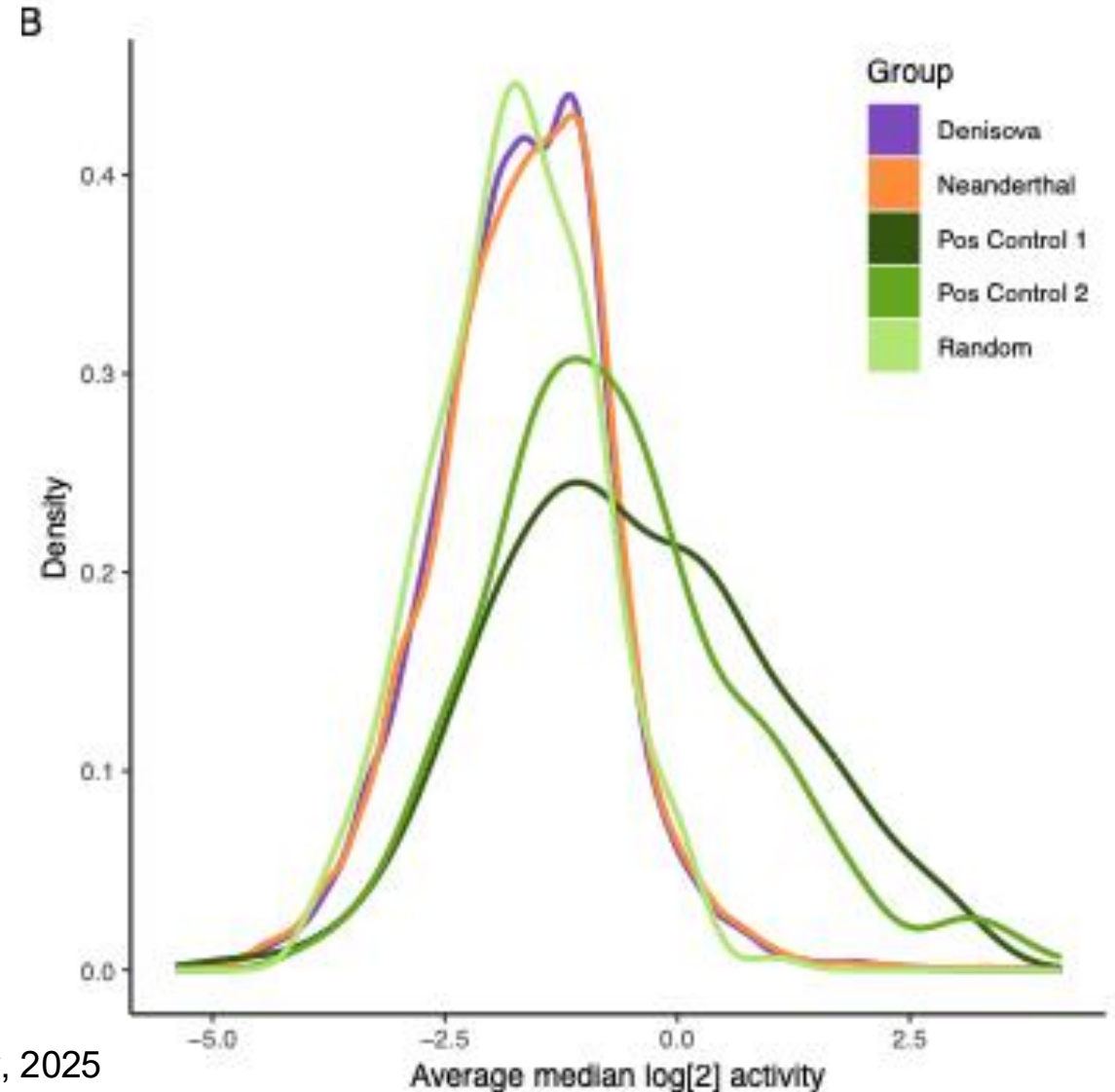
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- Many MPRA libraries do not
 - Depending on the design, most tested elements will be inactive



The challenge of defining activity

- Protein-coding DMS and SGE generally have inbuilt controls – wildtype sequence is expected to be 'active' and can be treated as baseline
- Many MPRA do not
 - Depending on the design, most tested elements will be inactive
- Defining external control sets is crucial for calibration and validation
 - Positive: validated cell-type specific enhancers? Known truth-sets?
 - Negative: scrambled sequences? Validated cell-type specific enhancers for other cell types?



Statistical data modelling (finally!):

- Most tools for handling MPRA data are R-based.
- Some options:
 - BCalm (Keukeleire et al, BMC Bioinformatics 2025), CRADLE (Kim et al, Genome Research 2021), STARRPeaker (Lee et al, Genome Biology, 2020), MPRAanalyze (Ashuach et al, Genome Biology 2019), mpralm (Myint et al, BMC Genomics 2019), QuASAR-MPRA (Kalita et al, Bioinformatics 2018)
- More general count-based methods for data modelling (from RNA-seq, come with their own assumptions):
 - DESeq2, limma/edgeR
- More artisanal (mostly deprecated):
 - One or two-step ANOVA designs

Coming soon: Best Practices recommendations!

- Collaborative manuscript and accompanying R bookdown textbook in prep
 - Better and more thorough examples of both good and bad libraries
 - Today's talk is a warm-up.

Best Practices for Massively Parallel Reporter Assays – Analyses, Diagnostics, and Interpretation

1. Introduction